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Safety

THE BIRD/WILDLIFE AIRCRAFT STRIKE HAZARD (BASH) REDUCTION PROGRAM

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This instruction provides a base program to minimize aircraft exposure to potentially hazardous wildlife strikes. It implements AFI 91-202, *US Air Force Mishap Prevention Program* and AFP 91-212, *Bird/Wildlife Aircraft Strike Hazard (BASH) Management Techniques.* This instruction applies to all host, associate, and TDY organizations on Travis AFB, including US Air Force Reserve members and units. The 60 AMW/CC is responsible for implementation of this instruction.

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SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include: (1) Clarifying and expanding BASH program Tasks and Responsibilities; (2) Further

refining exemptions to BASH Phase II restrictions; (3) Increasing detail for AHAS restrictions and waiver authorities when conducting low level operations; (4) Adding BASH guidance for Travis AFB crews operating at off-station locations; (5) Reorganizing chapter order and formatting for better information flow.

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Chapter 1

BIRD/WILDLIFE AIRCRAFT STRIKE HAZARD (BASH) PLAN

1.1. Situation.

1.1.1. General. Daily and heavy seasonal bird movements near Travis Air Force Base (Travis AFB) create various hazards to aircraft. This plan establishes a program to minimize the hazard from bird strikes as well as strikes with terrestrial animals. No single solution to the bird strike problem exists, and a variety of techniques and organizations are involved in the control program. This plan is designed to:

1.1.1.1. Establish a BHWG and designate responsibilities to its members.

1.1.1.2. Establish a Bird Hazard Committee (BHC) and designate responsibilities to its members.

1.1.1.3. Establish training for designated base members concerning responsibilities and actions.

1.1.1.4. Establish procedures to identify high hazard situations and to aid aircrews and supervisors in altering/discontinuing flying operations when required.

1.1.1.5. Establish aircraft and airfield operating procedures to avoid hazardous situations.

1.1.1.6. Provide for dissemination of information to all assigned and transient aircrews on specific bird hazards and procedures for avoidance.

1.1.1.7. Establish guidelines for activities to decrease the attractiveness of the airfield environment to birds and other animals IAW AFMAN 32-7003.

1.1.1.8. Provide guidelines for dispersing birds and moving wildlife when needed on and around the airfield.

1.1.1.9. Provide for coordination of efforts with other area airfields.

1.1.2. Airfield and Local Area.

1.1.2.1. The primary aircraft using the Travis AFB runway include the following base assigned aircraft: C-17, C-5 and KC-10. Other transient US Air Force, US Coast Guard, US Army, US Navy/Marines, foreign military aircraft and commercial cargo aircraft also use Travis AFB regularly.

1.1.2.2. Two passes through the hills west of Travis AFB act as major flight paths for birds flying inland from the coast. One of the passes is in close proximity to Travis AFB and is part of the Suisun Marsh, one of the last remaining natural wetlands in the area. Gulls, ducks, geese, and other birds leave and return to the marsh through this southwest pass en route to the Sacramento River Delta and lands surrounding Travis AFB.

1.1.2.3. The wildlife strike problem at Travis AFB is further complicated by numerous hunting clubs who maintain habitats to attract waterfowl for hunting and nearby landfills which attract large numbers of gulls. Two landfills are located on opposite ends of the Travis AFB runways. The Potrero Hill landfill in Fairfield is located 1.5 miles from the approach end of Runway 03L and the Recology Hay Road Landfill in Vacaville is 4 miles from the approach end of runway 21L. Considerable gull flight activity between the two sites increases the risk to aircraft.

1.1.2.4. Agricultural activities play a key role in maintaining large populations of small birds in the spring and early fall, as well as 400,000 to 500,000 blackbirds during the winter months. A large population of small mammals on the base and the surrounding agricultural land supports a significant raptor and coyote population.

1.1.2.5. Bird strikes with waterfowl are a serious problem at Travis AFB. This is due to Travis AFB being situated on the Pacific Flyway. The area around Travis AFB is also a wintering area for a very large population of ducks and a significant population of geese and swans. Migratory and daily flight activities by the birds create a substantial risk to flight operations. Ducks begin their daily flights just before sunrise and may continue to fly for up to two hours after sunrise. Morning duck movements begin east of Travis AFB with birds flying out of the Yolo Bypass, Hastings Slough, and the refuges north of Sacramento. Flocks flying east out of the Suisun Marsh and the Nurse, Denverton, and Hill Sloughs soon join these westerly flights. Low clouds, haze or fog often delays the departure of the birds, causing them to fly later in the day. Most late afternoon bird flights occur at sunset and may last up to an hour. Finally, small Neotropical migrant birds and local bat populations pose a risk to aircraft flying at night.

1.2. Execution.

1.2.1. Concept of Operations: Reducing the bird/wildlife strike hazard at Travis AFB requires a cooperative effort between many organizations, both on and off base. It requires multiple tactics, including habitat management, flight operation considerations, aircrew awareness, bird observation, and active and passive bird controls. The Office of Primary Responsibility (OPR) for coordination and execution of this plan is 60 Air Mobility Wing Flight Safety Office (60 AMW/SEF).

1.2.2. BHWG:

1.2.2.1. Function. The BHWG reviews data on bird strikes, recommends changes to operational procedures and habitat, and initiates changes to the 60 AMW BASH Plan.

1.2.2.2. Authority. The BHWG submits all major recommendations to the 60 AMW Commander (60 AMW/CC) for approval. Implementation of recommendations is through the normal chain of command.

1.2.2.3. Composition. 60 AMW/SEF, USDA Wildlife Services (USDA- WS) Biologists, Operations (flying squadrons, air traffic control (ATC), airfield management, and Operations Group Standardizations/Evaluations), Civil Engineers (natural resources, pest management, and as needed, operations, community planning, aircraft rescue and firefighting), and flying tenant units. The 60 Vice Wing Commander (60 AMW/CV) will chair the formal BHWG meetings. 60 AMW/SEF will act as the executive secretary for the group and at a minimum, maintain original meeting minutes. Judge Advocate, Public Affairs (60 PA) and Services will be forwarded the minutes from the BHWG for their review.

1.2.2.4. Meeting Schedule. The BHWG will meet at least semi-annually.

1.2.2.5. Meeting Agenda. The BHWG meetings will discuss, but are not limited to: local bird strike statistics, locally observed bird activity, local wildlife habitat management and modification, annual bird migration, local BASH Plan procedures and responsibilities, BASH awareness/education/training, flying schedule/bird activity conflicts, and changes to this BASH Plan. Additional topics may be discussed as needed.

1.2.3. Bird Hazard Committee (BHC):

1.2.3.1. Function. The BHC is an Action Officer (AO) level group that meets to discuss BASH ideas and issues in an informal round-table setting. Issues are socialized among the different agencies with the goal of a cohesive agenda and action plan ready to present to the BHWG.

1.2.3.2. Composition. 60 AMW/SEF is the AO for these meetings. The group will primarily consist of 60 AMW/SEF, USDA-WS Biologists, 60 Operations Support Squadron (60 OSS), 60 Civil Engineering Squadron (60 CES) and 60 Operations Group (60 OG). Other members, to include squadron Flight Safety Officers, may be asked to attend based on the meeting agenda.

1.2.3.3. Meeting Schedule. The BHC will meet at Quarterly Bird Mitigation Meetings between BHWG meetings, with a goal of meeting every three months. Committee members are encouraged to use electronic means to keep the group updated as action items progress or change to facilitate more productive meetings.

1.2.3.4. Meeting Agenda. The BHC will discuss the same topics as the BHWG and will also discuss the agenda to present to the BHWG.

Chapter 2

TASKS AND RESPONSIBILITIES

2.1. Vice Wing Commander (60 AMW/CV)

2.1.1. Will chair and be the approval authority for recommendations of the BHWG.

2.2. Flight Safety (60 AMW/SEF)

2.2.1. Acts as the OPR for the content, execution and compliance of this plan.

2.2.2. Will collaborate with California Department of Fish and Wildlife (CDFW), U.S. Fish and Wildlife Service (USFWS), USDA-WS, and HQ AFSEC/SEFW to stay abreast of local wildlife populations, movements, and control techniques.

2.2.3. In conjunction with 60 CES, Airfield Management, and USDA-WS, 60 AMW/SEF will inspect the airfield at least quarterly for any conditions that may attract wildlife.

2.2.4. 60 AMW/SEF is the designated OPR for maintaining all applicable federal and state depredation permits or Migratory Bird Treaty Act (MBTA) permits and associated documents when those permits are in support of the BASH program.

2.2.5. Establishes a bird strike hazard awareness program in conjunction with 60 OG/OGV, to include information on local bird hazards and reporting procedures.

2.2.6. Monitors base wide compliance and reports all aircraft bird strikes and hazards per AFIs 91-202 and 91- 204.

2.2.7. Coordinates with aircrew and maintenance personnel for collection of bird remains after bird strikes. Forwards wildlife remains for identification IAW AFI 91-212 and AFI 91-202.

2.2.8. Establish and review BASH plan annually for compliance with current directives and changing conditions/environment.

2.2.9. Will notify 60 OG/CC when they recommend implementing or terminating Phase II restrictions.

2.2.10. Will procure and maintain at least two BASH 870 shotguns for the organizations responsible for wildlife depredation.

2.2.11. Will designate 60 AMW/SEF Non-commissioned Officer (FSNCO) or Flight Safety Officer (FSO) as the BASH Program Manager.

2.2.12. Will track all BASH Phase II window violation/deviations for trend analysis.

2.3. BASH Program Manager

2.3.1. Will be a member of 60 AMW/SEF, typically the Flight Safety Non-Commissioned Officer (FSNCO).

2.3.2. Seeks and obligates funding for routine BASH-related operations and minor projects.

2.3.3. Notifies 60 CES/CEIE of any adverse trends or effects to natural resources related to BASH activities and coordinates with CES to bring broader wildlife management resources to bear when appropriate.

2.3.4. Organizes BHWG meetings, invites members to attend, and produces minutes.

2.3.5. Organizes Quarterly Bird Mitigation Meetings, invites members to attend, and produces minutes.

2.3.6. Ensures base wide compliance of BASH Hazard Mitigation (**Chapter 4**).

2.3.7. Will enforce compliance of WEZ.

2.3.8. Will designate a Deputy BASH Program Manager, which will normally be the USDA-WS Biologists.

2.3.9. Will coordinate with Pest Management Shop for rodent and insect control and nest removals.

2.3.10. Will consult with the Travis AFB Natural Resource Manager on natural resources management issues and strategies.

2.3.11. Will coordinate Mission-Essential Personnel (MEP) status for USDA-WS personnel and the FSNCO (See **Attachment 2**).

2.3.12. Maintains contact with Potrero Hills landfill personnel and any other off base locations deemed appropriate for any Travis AFB BASH related issues.

2.3.13. Authorized USDA-WS Personnel will have a signed letter from 60 AMW/SEF to 60 Security Forces Squadron (60 SFS) for use of firearms on the installation.

2.4. Deputy BASH Program Manager – USDA – WS

2.4.1. At the request of the airfield manager, provides training on bird dispersal techniques.

2.4.2. Oversees all harassment and depredation activities on the airfield.

2.4.3. Presents BASH updates at the Quarterly Flight Safety meetings (QFSM).

2.4.4. Monitors bird activity and maintains bird strike statistics of local and transient aircraft.

2.4.5. Provides wildlife identification training for personnel conducting flight line wildlife abatement.

2.4.6. Provides monthly activity reports to 60 AMW/SEF and other interested organizations.

2.4.7. Monitors habitat on and around the WEZ for attractants to wildlife.

2.4.8. Monitors bird/wildlife activity on the flight line and makes recommendations on the BWC.

2.4.9. Authorized to conduct wildlife activities as coordinated and approved under applicable depredation permits and Travis AFB Integrated Natural Resource Management Plan (INRMP). These activities may entail but are not limited to, the use of the following tools and techniques:

2.4.9.1. Structural modifications including netting and spikes

2.4.9.2. Pyrotechnics (bird whistlers, bird bombs, shell crackers and CAPA charges)

2.4.9.3. Traps (snares, conibears, Goshawk, padded foothold cage traps)

2.4.9.4. Firearms (air rifle, rifle, and shotgun)

2.4.10. Wildlife abatement activities will primarily be conducted in and around the WEZ (includes the runway, taxiways, parking aprons, and hangars).

2.4.11. Will inspect the airfield perimeter security fence at least monthly and after a significant weather event to detect and report breaches that allow wildlife to enter the airfield environment.

2.4.12. Will be granted access and authority to conduct wildlife hazard management operations within all lands owned by the installation, that directly affect mission capability and flight safety.

2.4.13. Will be issued restricted area badges in order to have access to restricted flight line areas and hangars.

2.4.14. Authorized to maintain a Licensed Explosive Facility on Travis AFB IAW AFMAN 91-201 for the storage of ammunition, explosive charges and other munitions strictly necessary to accomplish BASH program requirements.

2.4.15. Authorized to store, IAW AFI 31-101 and use, on Travis AFB, those firearms strictly necessary to the conduct of BASH program requirements

2.5. Command Post

2.5.1. Notify inbound aircraft that initiate radio contact of any BWC of MODERATE or SEVERE. Include location, movement, and other known data of the bird activity that is driving the BWC.

2.5.2. After being notified of a new BWC from Airfield Management, broadcast the new BWC over VHF/UHF radio frequencies, and advise aircrews of the BASH Phase II windows (if applicable). In addition, Command Post will immediately notify the OG/CC any time the BWC increases or is downgraded. Command Post will broadcast the change in BWC to all aircraft.

2.5.3. Ensure transient aircraft involved in a wildlife strike fill out an AF Form 853, Air Force Wildlife Strike Report, sample remains and forward to Wing Safety.

2.5.4. Disapprove any request, except those exempt, from transient crews to conduct transition training, as well as any requests to schedule a takeoff or landing when Phase II restrictions are in effect. Ensure all aircraft, both local and transient, comply with flight restrictions listed in this instruction.

2.5.5. Coordinate delays, diversions, and release of AMC controlled aircraft based on BWC. Issue bird watch advisories to all aircraft arriving and departing Travis AFB. Notify 618th Air Operations Center, Tanker Airlift Control Center (TACC) when wildlife conditions could result in possible aircraft diversions or delays.

2.5.6. Assign X113 delay code in GDSS2 for all AMC missions delayed due to wildlife hazard conditions.

2.6. Air Traffic Control (ATC)

2.6.1. Monitor level of bird activity using ATC observation, aircrew reports, and wildlife control reports.

2.6.2. Notify Airfield Management and/or wildlife dispersal personnel directly whenever wildlife activity on the airfield requires action.

2.6.3. ATC watch supervisors can raise the BWC based on wildlife activity. ATC watch supervisor can forward downgrade requests of the BWC to Airfield Management if visual observation of bird activity reveals activity is not a probable hazard to flying safety. After a reported bird strike, the BWC may be raised to MODERATE as necessary. At a minimum, ATC supervisor will ensure a reassessment of the BWC occurs.

2.6.4. ATC will update the BWC in the Airfield Automation System (AFAS) when notified by designated personnel.

2.6.5. ATC will notify Airfield Management and RAPCON when the BWC is changed. Airfield Management will notify Command Post.

2.6.6. ATC will broadcast the BWC on the Automated Terminal Information Service (ATIS).

2.6.7. The BWC shall be included on the ATIS when the BWC is MODERATE or SEVERE, regardless of Phase I or II.

2.6.8. When the BWC is rapidly changing, ATC will include the following statement on the ATIS: “Due to rapidly changing bird watch conditions, contact Travis ATC or Travis Approach for current bird watch condition.”

2.6.9. During BASH Phase II, ATIS will broadcast the effective local times of the Phase II windows.

2.6.10. Coordinate activities with wildlife dispersal units.

2.7. RAPCON

2.7.1. Traffic conditions permitting, RAPCON will use the northern radar traffic pattern or a minimum radar traffic pattern altitude of 3,000 feet to the south when a bird hazard exists at low altitude over the wetlands south of the runways.

2.8. Airfield Management

2.8.1. Coordinate with wildlife dispersal personnel for assistance with wildlife dispersal, depredation and trainings.

2.8.2. Notify ATC of hazardous bird/wildlife activity on or around airfield, including BWC downgrade requests.

2.8.3. Notify Command Post of BWC updates.

2.8.4. Report known wildlife strikes to 60 AMW/SEF.

2.8.5. Retrieve carcasses on taxiways and runways or, if necessary, contact USDA–WS Personnel for assistance.

2.8.6. Maintain any bio-acoustic and pyrotechnic equipment necessary for bird control. Personnel assigned will maintain proficiency on above devices. Only stock-listed, USAF approved munitions will be procured or utilized on the airfield. Munitions must be ordered through 60th Maintenance Squadron Munitions Flight. Munitions will be stored in approved areas in quantities not to exceed a projected 60-day supply IAW Wing Weapons Safety Guidance. An explosive facility license (AF form 2047) will be current and posted on the storage locker door or side panel.

2.8.7. During Phase II whenever present on the airfield, Airfield Management personnel will carry wildlife harassment tools.

2.8.8. Identifies and monitors wildlife habitat attractants such as grass height, standing water drainage, foliage and vegetation control. Requests management execution with PRIDE personnel and 60 AMW Civil Engineering.

2.8.9. Airfield Management will be responsible for notifying all aircrews (military and civilian) of the BWC and BASH window when the aircrews file their flight plan.

2.8.10. Coordinate with USDA-WS personnel to evaluate and confirm a BWC downgrade upon the request of the ATC watch supervisor, if visual observation of bird activity reveals activity is not a probable hazard to flying safety.

2.8.11. Publish a NOTAM identifying Phase II windows and restrictions. This will allow commercial and non-DOD aircraft who don't have access to AP1 to be aware of Phase II windows and restrictions. The NOTAM will include a warning to non-exempt aircraft (i.e. general aviation flying operations) that the airspace is considered hazardous due to elevated bird activity and may be denied access.

2.8.12. Will report all BASH Phase II window violations to 60 AMW/SEF. Violation reports will be reported within 24 hours of the event.

2.9. Operations Group Commander (OG/CC)

2.9.1. The 60 OG/CC will implement and terminate Phase II upon notification from 60 AMW/SEF of bird activity that has increased or decreased significantly.

2.9.2. 60 OG/CC will make operational changes to avoid areas and times of known hazardous bird concentrations, mission permitting.

2.9.3. The 60 OG/CC will approve the total number of scheduled missions (training and operational) which fall within the BASH windows.

2.10. Operations Group Standardization/Evaluation

2.10.1. Establish procedures for aircrew following a known or suspected bird/wildlife strike. Crews should conduct a complete inspection of the aircraft to include all engine inlets, engine fan blades, cooling air inlets, pitot/static inlets, and flight controls will be inspected for bird strike damage, regardless of the location of the known bird strike. Guidelines will also require aircrews to complete an AF Form 853 (Air Force Wildlife Strike Report) and AMC Form 97.

2.10.2. Disseminate BASH procedures for aircrew with the respective weapons systems regulations (OI 11-2MDSV3) and Flight Crew Information File/Books (FCIFs/FCBs). BASH knowledge will be continually evaluated and appropriate risk-reducing changes for aircrew flight operations will be disseminated via FCIF directives.

2.11. Civil Engineer Squadron

2.11.1. Monitor grass mower contractor compliance with the grass management program.

2.11.2. Natural Resources will document depredation activity.

2.11.3. Conduct pest management including insect and rodent control.

2.11.4. Complete submitted work orders in an expeditious manner to correct environmental conditions that increase BASH potential.

2.11.5. Comply with habitat management. Seasonally inspect the "clear zone" areas off the approach end of runways 21L/3R, 21R/3L and 21 Assault/3 Assault for grain type crops, vegetation, or industrial development attracting wildlife. Maintain vegetation within the Aircraft Movement Area in accordance with AFI 91-212 paragraph 3.2.1.2. If a contractor performs grass cutting, monitor performance to ensure compliance with this instruction. When necessary and legal to do so, eliminate foliage in drainage ditches. As a last resort, consult with the Water Board for the best course of action in dealing with large vernal pools that attract waterfowl.

2.11.6. Maintain Drainage Ditches. Regularly inspects ditches to keep them clear of vegetation and debris. Maintain ditch sides as steep as possible (IAW AFI 91-212) and mow vegetation to discourage waterfowl, wading birds and emergent vegetation. Improve drainage as necessary to control temporary ponds or puddles.

2.11.7. Prohibit Wildlife Feeding. Signs should be placed in picnic areas to educate the public on the hazard posed by feeding wildlife, particularly waterfowl.

2.12. Current Operations

2.12.1. Enter BASH Phase I and Phase II start/end dates (normally BASH Phase II is in effect from 1 October - 30 November & 1 February - 30 April) into GDSS2 Airfield Data Base as a planning remark in the Temporary Restrictions/Planning Remarks section.

2.12.2. Schedule takeoffs/arrivals to comply with BASH Phase II restrictions.

2.12.3. Coordinate with TACC to schedule mission departures to comply with BASH Phase II restrictions to the maximum extent possible. Note: Missions will not use on time take off criteria (e.g. -20/+14 minutes) to depart inside the BASH Phase II window.

2.12.4. Enter sunrise, sunset, and BASH windows on the daily schedule.

2.13. Maintenance Group

2.13.1. Establish procedures for responding when an aircraft is known or discovered to have a bird strike. Procedures will include at a minimum:

2.13.2. Determine inspection requirements for all engine inlets/exhaust, engine fan blades, cooling air inlets, pitot/static inlets, and flight controls.

2.13.3. When a bird strike is discovered, collect all bird remains found to include whole birds, beak, feet, feathers, and blood smear remains (SEE Attachment 4). Bag remains, taking care to separate and label based on individual strike location. Complete the entire AF Form 853 Air Force Wildlife Strike Report (SEE Attachment 5) and if sections are not applicable, please indicate so. Provide labeled remains and completed AF Form 853 to 60 AMW/SEF within 48 hours of discovering the strike.

2.13.4. If considering impounding an aircraft for bird strike damage for investigation, notify Command Post and 60 AMW/SEF immediately.

2.14. Security Forces (60 SFS)

2.14.1. Provide training to 60 AMW/SEF and Airfield Management to maintain currency in use of shotguns IAW AFI 36-2654, *Combat Arms Program*.

2.14.2. 60 SFS will only certify Air Force BASH team members.

2.15. Public Affairs (60 PA)

2.15.1. Participate as required to provide a public information program designed to inform base personnel, dependents, and the general public of the hazards and costs of uncontrolled bird/wildlife activities and the measures being taken to minimize them. This information will educate the base on the hazard that wildlife poses to aircraft operations and efforts required by all personnel and residents to prevent bird strikes.

2.15.2. Approval. Release authority for all activity information in this plan rests with the 60 AMW/CC.

2.15.3. Assist in the preparation of educational information for the base community. This may consist of the following actions:

2.15.4. Providing imagery and information to be included in distributed publications such as the Travis Tailwinds newsletter.

2.15.5. Drafting media messages to stress the need for vigilance by all base personnel in bird detection efforts. Additionally, these messages should reiterate the strict “no feeding wildlife” policy.

Chapter 3

BASH PROGRAM OPERATIONAL INFORMATION

3.1. General

3.1.1. This chapter outlines specific procedures and restrictions during hazardous bird conditions. An advisory will be published in the DoD Flight Information Publication AP/1 highlighting the general and seasonal bird hazards on the airfield and in the local flying area.

3.2. Bird Watch Condition

3.2.1. IAW DoD Flight Information Handbook, AFI 91-202, and AFI 91-212 AMC Supplement, the following terminology has been established for rapid communication of bird activity. Bird conditions should be relayed with the BWC code. Although BWC is frequently defined as a certain number of birds, Travis AFB refrains from this due to our unique runway configuration and wildlife present.

3.2.2. SEVERE: Wildlife activity on or immediately above the active runway or other specific location representing high potential for strikes. Supervision and aircrews must thoroughly evaluate mission need before conducting operations in areas under condition SEVERE.

3.2.3. MODERATE: Wildlife activity near the active runway or other specific location representing increased potential for strikes. BWC MODERATE requires increased vigilance by all agencies and supervisors and caution by aircrews. At Travis AFB, this condition exists when concentrations of birds are observed in the WEZ. Positive actions will be taken by all agencies to disperse the concentrations of birds that are causing the hazard. Bird concentration areas will be tracked for future habitat modification.

3.2.4. LOW: Wildlife activity on and around the airfield representing low potential for strikes.

3.2.5. **NOTE:** BWC is a reactive tool based on observations of local airfield wildlife activity and are independent of Bird Avoidance Model (BAM) or Avian Hazard Advisory System (AHAS) risk hazard levels.

3.3. Declaration of Authority

3.3.1. Aircrew, 60 AMW/SEF, USDA-WS Personnel, 60 OSS/OSAA or other operations personnel will call ATC to report observed bird activity until it has dispersed.

3.3.2. Personnel authorized to adjust the BWC:

3.3.2.1. Increasing the BWC:

3.3.2.1.1. 60 OSS/OSAA

3.3.2.1.2. 60 AMW/SEF

3.3.2.1.3. USDA-WS Personnel

3.3.2.1.4. ATC Watch Supervisor

3.3.2.2. Decreasing the BWC:

3.3.2.2.1. 60 OG/CC

3.3.2.2.2. 60 OSS/OSAA

3.3.2.2.3. 60 AMW/SEF

3.3.2.2.4. USDA-WS Personnel

3.4. Notification

3.4.1. Ground Personnel. If a bird hazard exists, notify airfield management as applicable. This notification can be made on a radio network or by telephone. Airfield management monitors the Ramp network. The report should include:

3.4.1.1. Caller identity (agency for ground personnel, call sign for aircrews)

3.4.1.2. Location

3.4.1.3. Altitude

3.4.1.4. Time of sighting

3.4.1.5. Approximate number of birds

3.4.1.6. Type of birds (if known)

3.4.1.7. Behavior of birds (soaring, flying to or from a location)

3.4.2. Aircrews. Report significant bird activity as follows:

3.4.2.1. At Travis, or any other local training airfield - notify ATC, or airfield management

3.5. BWC Dissemination

3.5.1. The Declaration of a BWC starts a process of notification and action. For the directed actions associated with MODERATE or SEVERE conditions, see **paragraphs 3.7.1.2** and **3.7.1.3** Disseminate updated BWC's as follows:

3.5.2. ATC:

3.5.2.1. Notifies departing and arriving aircraft of BWC.

3.5.2.2. Passes BWC to Airfield Management.

3.5.2.3. Update the ATIS to reflect BWC of MODERATE or SEVERE. For rapidly changing BWC, place a statement on ATIS advising aircrews to contact Ground, ATC.

3.5.3. Airfield Management:

3.5.3.1. Coordinates with ATC to upgrade or downgrade the BWC.

3.5.3.2. Airfield Management Operations will post the BWC on the status board in Airfield Management.

3.5.3.3. Dispatches personnel to the site of the bird hazard to conduct harassment or monitor status of hazard.

3.6. BASH Phases

3.6.1. PHASE I extends from 1 May to 30 Sep and 1 Dec to 31 Jan and is typically the period of lower threat of bird strike.

3.6.2. PHASE II extends from 1 Oct to 30 Nov and 1 Feb to 30 April and is the period of increased bird concentrations resulting in an elevated risk of bird strikes. Operational restrictions are implemented to mitigate the additional risk. Phase II will be implemented by the 60 OG/CC upon recommendation by USDA-WS / 60 AMW/SEF. 60 OSS/OSAA will publish a NOTAM identifying Phase II windows and restrictions.

3.6.3. These periods are published in the AP1 but can be shifted when significant wildlife activity occurs earlier/later than the dates listed above.

3.6.4. NOTE: BASH Phases II is a proactive scheduling tool based on historical wildlife activity, bird strike data and regional migration patterns.

3.7. Flight Restrictions

3.7.1. BWC Operational Restrictions:

3.7.1.1. All Touch-and-Go Landings are prohibited during BWC MODERATE and SEVERE with no exemptions.

3.7.1.1.1. LOW Operational Restrictions:

3.7.1.1.2. There are no operational restrictions outside of BASH Phase II.

3.7.1.2. MODERATE Operational Restrictions:

3.7.1.2.1. Initial takeoff and full stop landings will be allowed only when departure and arrival routes avoid identified bird activity.

3.7.1.2.2. All local IFR/VFR traffic pattern activity will cease. Airborne aircraft will divert, hold, or land. When making this decision, Aircraft Commanders will use all available means to ensure arrival/departure routes avoid known bird activity.

3.7.1.2.3. Airfield Management, Wing Safety or USDA-WS Personnel will survey the runways/airfield environment for significant bird activity prior to any aircraft activity.

3.7.1.3. SEVERE Operational Restrictions:

3.7.1.3.1. All DoD and non-exempt flight operations (takeoffs, landings and approaches) are prohibited without BWC SEVERE waiver. 60 OG/CC, 60 AMW/CV, or 60 AMW Wing Commander (60 AMW/CC) are the approval authorities.

3.7.1.3.2. Airborne DoD and non-exempt aircraft will divert or hold.

3.7.1.3.3. 60 OSS/OSAA, 60 AMW/SEF or USDA-WS Personnel will survey the runways/airfield environment for significant bird activity prior to any aircraft activity.

3.7.2. BASH Phase II Operational Restrictions:

3.7.2.1. Once the BASH Phase II window is implemented by the 60 OG/CC, The BASH Phase II Window is in effect from 30 minutes before to 1 hour after sunrise and 1 hour before to 30 minutes after to sunset.

3.7.2.2. To the maximum extent possible, all operational and training missions, to include waiver exempt aircraft, should not be scheduled to depart/arrive within the BASH windows.

3.7.2.3. EXCEPTION: Unscheduled or delayed aircraft are allowed one takeoff OR one full-stop landing if the BWC is reported as LOW. Scheduled aircraft attempting to takeoff early within the window will require a Conditional Waiver regardless of BWC.

3.7.2.4. All DoD aircraft will adhere to the restrictions during declared BASH windows; operations violating these restrictions by DoD aircraft require an OG/CC conditional waiver.

3.7.2.5. All Touch-and-Go Landings are prohibited during BASH Phase II windows with no exemptions.

3.7.2.6. Published NOTAM identifies Phase II windows and restrictions. This allows commercial and non-DOD aircraft who don't have access to AP1 to be aware of Phase II windows and restrictions. The NOTAM will include a warning to non-exempt aircraft (ie. General aviation flying operations) that the airspace is considered hazardous due to elevated bird activity and may be denied access.

3.7.2.7. The total number of scheduled missions (training and operational) which fall within the BASH windows will be approved by 60 OG/CC and annotated in the monthly 60 Wing Operations Plan (60 OSS/OSO). Additionally, all missions will be reviewed and approved bi-weekly at the Air Operations Directive meeting.

3.8. BASH Phase II/BWC Exemptions.

3.8.1. Exempt aircraft do not require a BWC/BASH window conditional waiver but are highly encouraged to modify their arrival or departure times to avoid BASH Phase II windows. These aircraft include:

3.8.1.1. Emergency aircraft

3.8.1.2. Commercial carriers

3.8.1.3. Foreign military aircraft

3.8.1.4. Life Flight aircraft

3.8.1.5. TACAMO operations

3.9. Phase II Window Violations/Deviations

3.9.1. Any deviations or violations of the BASH Phase II window restrictions will be reported to and tracked by 60 AMW/SEF (See **Attachment 3**).

3.10. Waiver Process

3.10.1. The procedure to coordinate BASH Phase II or BWC conditional waivers is as follows:

3.10.2. Non-exempt Aircrews will:

3.10.2.1. Contact 60 AMW/CP via radio (VHF 141.9, UHF 349.4) prior to engine start or NLT 100 NM prior to arrival to allow adequate time for approval.

3.10.2.2. Request an OG/CC Conditional Waiver approval through the 60 AMW/CP or validate that a previously approved waiver is still valid.

3.10.3. ATC will:

3.10.3.1. Upon initial contact, instruct aircrew to contact 60 AMW/CP to begin waiver process.

3.10.3.2. **NOTE:** ATC should refrain from soliciting an aircrew's waiver status beyond conveying approval or denial, waiver coordination is between the aircrew and 60 AMW/CP.

3.10.3.3. If needed, pending 60 OG/CC waiver coordination, direct aircraft to hold. If waiver is denied, ask the aircrew if they wish to continue to hold or if they wish to divert to another airfield.

3.10.3.4. ATC will contact 60 OSS/OSAA for the latest BWC assessment and relay it to aircrew.

3.10.3.5. In the event of an aircraft in-flight emergency or emergency fuel, the aircraft will be permitted to land, inform the aircrew of the current BASH conditions.

3.10.3.6. **NOTE:** Aircraft declaring minimum fuel will continue to be instructed to contact 60 AMW/CP to request a conditional waiver.

3.10.4. 60 AMW/CP will:

3.10.4.1. Contact 60 OG/CC to request a Conditional Waiver.

3.10.4.2. Document approval note into GDSS2 for BASH Phase II waivers.

3.10.4.3. Contact ATC to advise approval/denial of Conditional Waiver.

3.10.4.4. Waiver Request Denial: Relay denial to aircrew. 60 AMW/CP will also advise ATC that the waiver has been denied and that the aircraft will not be permitted to take-off or land.

3.10.4.5. Waiver Request Approval: Relay approval to aircrew. 60 AMW/CP will also advise ATC that the waiver has been approved and that the aircraft will be permitted to take-off or land.

3.11. Mission Planning

3.11.1. Consult AFMAN 11-202V3, for mission planning requirements. Aircrew briefings on bird strikes and other wildlife hazards will include, as a minimum:

3.11.1.1. Potential bird hazards and attractants along the proposed route of flight using the Bird Avoidance Model (BAM) and the Avian Hazard Advisory System (AHAS). Further information about the BAM and AHAS is found below and can be accessed on the Internet.

3.11.1.2. Actions, if flocks of birds are encountered.

3.11.1.3. Mission abort options/requirements due to a Bird/wildlife activity or strike.

3.12. Bird Avoidance Model

3.12.1. This historically-based predictive model uses GIS technology as a tool for analysis and correlation of bird habitat, migration, and breeding characteristics, combined with key environmental and man-made geospatial data. The Bird Avoidance Model was primarily designed to provide schedulers and biologists with a long-range avian prediction model. It can be accessed at **www.usahas.com** (Select Area, then Content Display Google Map). The model is extremely useful in depicting favorable avian geographic locations/features and other secondary avian prediction aids (such as Next Generation Weather Radar (NEXRAD) and previous bird strike locations) along a planned flight route for situational awareness.

3.13. Avian Hazard Advisory System

3.13.1. AHAS utilizes the Bird Avoidance Model as a foundation supplemented by NEXRAD weather radar to provide a near real-time description of bird activity. The system may be accessed at **www.usahas.com**. It does not make recommendations on restrictions to be imposed for any category of bird strike risk (Low, Moderate, or Severe). These decisions will be made at the local level after careful consideration of the mission design series, mission, and mission priority. The web site contains a continually revised tutorial explaining its development and use as well as downloadable training slides.

3.14. AHAS Restrictions:

3.14.1. AHAS Risk LOW: No restrictions.

3.14.2. AHAS Risk MODERATE: On Military Training Routes (MTRs) crews will fly no faster than 250 knots OR no lower than 1,000 feet AGL on affected route segments. Crews may request a waiver to fly MODERATE portions of low-levels with no restrictions from the SQ/CC (delegable to SQ/DO) based on training requirements. All wavier approvals will be sent to 60 AMW/SEF (**60AMW.SEF@us.af.mil**) for tracking purposes.

3.14.3. AHAS Risk SEVERE: On MTRs crews will fly no faster than 250 knots AND no lower than 1,000 feet AGL (3,000 feet AGL at night) on affected route segments. Also, the aircraft commander will obtain 60 OG/CC or 349 OG/CC approvals (as appropriate for the crew) prior to commencing any flight on affected route segments identified as AHAS Risk SEVERE. Approvals will be sent to 60 AMW/SEF (**60AMW.SEF@us.af.mil**) for tracking purposes.

3.14.4. In all cases, operational mission priority must be weighed in determining waiver approval. When a waiver is approved for operation during BWC SEVERE at AMC locations, the local Operations Group Commander (or higher) must actively monitor launch and recovery of aircraft.

3.15. Operational Modifications

3.15.1. The key to reducing Bird/wildlife strikes when modifying flight operations is to avoid known bird attractants, concentrations or movements of birds/wildlife. The following will help reduce wildlife hazards by modifying operational procedures:

3.15.1.1. Aircraft making formation departures increase the risk of damage from bird strikes when birds are feeding, loafing, or traversing near the runway. Formation and single-ship interval takeoffs often result in birds being scared up by the lead aircraft, causing the wingman to impact the birds. When the lead aircraft scares up large flocks of birds, the wingman should delay departure until the birds are clear of the runway. Pilots of lead aircraft are responsible for alerting wingmen of observed bird hazards during takeoff.

3.15.1.2. When flocks of migratory birds are observed, formation takeoffs and single-ship interval takeoffs with minimum spacing involving rejoins, increases the risk of serious bird strikes. The increased speed required to rejoin the lead aircraft after takeoff increases the risk of damage from bird strikes. When birds are known to be flying in the area, departures under Visual Meteorological Conditions (VMC) should be modified to reduce risk. Departures should be made in trail, with the rejoin beginning after the aircraft passes 2,000 to 3,000' Above Ground Level (AGL). If aircraft are to enter a low-level route immediately or stay at an intermediate altitude for a prolonged period, tactical formation normally provides enough aircraft clearance to allow wingmen to stay clear of birds.

3.15.2. Aircrews experiencing en route bird strikes should abort the mission when practical. **(T-3)** While some engine ingestions or a windscreen strike may be readily apparent from the flight deck, the damage from many engine, fuselage, and wing, tail, or random strikes cannot be adequately assessed inflight. Continuing a mission may cause greater structural or engine damage and lead to a serious emergency.

3.15.3. When sighting wildlife, aircrews will notify the controlling ATC facility and other aircrews if mission requirements allow.

3.16. Low-level Operations

3.16.1. Flying low-level routes or low-level operations in special use airspace involves higher aircraft speeds and greater exposure within bird flight environments. Operations in this flight regime invariably results in an increase in the number of damaging bird strikes. During these flights, aircrews are typically involved in cockpit duties, allowing little time to monitor bird activity. Visually maintain an active scan of the local environment as much as possible during critical operations. Consider the following when operating aircraft in the low-level environment: **(T-3)**

3.16.1.1. At a minimum, AHAS will be checked prior to mission execution. In an effort to reduce the bird strike hazard along low- level routes, crews will observe the following restrictions on training missions

3.16.1.2. Avoid areas with known raptor (birds of prey) concentrations during summer, between 1000 and 1700 hours due to increased thermals. Generally, a maximum altitude of 3,000-4,000' above ground level is reached by all raptor species, though soaring can occur at considerably higher altitudes. Areas with ideal terrain for creating thermals during summer months include ridgelines and rolling hills.

3.16.1.3. Unless mission essential, avoid flying one hour before and after sunrise/sunset to reduce potential hazards when there is a known increase in Bird/wildlife activity.

3.16.1.4. Avoid altitudes with the most favorable wind speed and direction for migrating birds (particularly near shear altitudes) up to 48 hours prior to and 24 hours after frontal passage; especially during October and November. Prevailing weather is a prime stimulus for migratory bird movements

3.16.1.5. Avoid flying near or over wildlife refuges, estuaries, parallel along shorelines, river corridors, landfills, stockyards, food processing plants or other known significant bird attractants below 3,000' above ground level.

3.17. Travis Aircrew Off-station BASH Procedures.

3.17.1. When operating at non-AMC airfields, aircrews will use the BWC definitions in **Paragraph 3.2** to assess the bird hazard and apply the operational restrictions in **Paragraph 3.7** Prior to transitioning at non-AMC airfields, planners/aircrews will obtain any available information on anticipated/actual bird activity at the airfield. Aircrews experiencing significant wildlife activity at these locations will advise their respective safety office BASH representative. NOTE: The 618th Tanker Airlift Control Center (TACC) or designee has approval authority for AMC-tasked flight operations at non-AMC locations.

3.17.2. Aircraft commanders will follow BWC-moderate procedures if they receive a civilian ATC/ATIS hazard advisory to use caution for birds in the vicinity IAW AFMAN 11-202V3.

3.18. Local Transition Hazards

3.18.1. The purpose of this section is to highlight both general and specific wildlife hazard information for local transition airfields.

3.18.2. Before initial approach, aircraft commanders will check the current wildlife hazard conditions. If information is not available (especially during Phase II), pilot discretion should be used before conducting the approach at the airfield. Airfields listed below may be in Phase II at the same period as Travis, and Operational Phase II restrictions would then apply.

3.18.2.1. Beale AFB (KBAB): During the Phase II period, a significant wildlife hazard exists. Large rice fields are located on the approach end of runway 33, providing winter nesting areas for waterfowl (swans, raptors). Local hunting clubs raise a large population of pheasant in the same area. The airfield area has a large number of meadowlarks and horn larks during the Phase II period. The base also has a significant coyote population that can pose a year round hazard to operations. USDA-WS have employees working at KBAB to reduce the threats wildlife cause to aircraft operations. Contact base operations at DSN 368-2002 for current conditions.

3.18.2.2. Sacramento International Airport (KSMF). Extensive agricultural areas (rice and corn) and brush lands that are highly attractive to many species surround the airfield. The airfield is also surrounded by waterways, including the Yolo Bypass, Tule Canal, Sacramento River, the irrigation canal maze, and Teal Bend Golf Course. Civilian airports do not identify daily "Bird Watch Conditions". However, the FAA provides information about civilian airports in the quarterly publication of the Airport/Facility Directory-Southwest U.S. Pilots should consult NOTAMS and the ATC for the latest wildlife information. KSMF Wildlife Mitigation Personnel reduce the threats wildlife cause to aircraft operations. Do not conduct transition training at KSMF when Travis AFB is in BWC-Moderate or above IAW OI11-2MDSV3.

3.18.2.3. Stockton Metropolitan Airport (KSCK). USDA–WS Personnel have not surveyed this airfield recently but it does have a landfill approximately two miles south of runway 29. The landfill turns the soil to reduce bird activity but caution should be used to avoid this area. In addition, wetlands are located south of the field. ATC may be contacted for current wildlife activity at (209) 982-9541.

3.19. Travis Aero Club Operations

3.19.1. The purpose of this section addresses the Bird/Wildlife Aircraft Strike Hazards affecting the Travis Aero Club.

3.19.2. The Travis Aero Club operates from a civilian airfield at Rio Vista Municipal Airport, CA (O88). Bird/Wildlife Aircraft Strike Hazards exist to a lesser degree for Aero Club aircraft due to their lower airspeeds, reciprocating engines, and mission operations.

3.19.3. While the strike risk to slow moving aircraft is lower, a significant bird hazard exists at O88 where bird strikes with waterfowl are still highly probable. The airport is bordered by the Sacramento River to the east and several canals surrounding the airport to the north, east and south. The canals cross the approach/departure path within close proximity of both runways. The strike problem is further complicated by large amounts of inland waterways to the east and southeast of the Sacramento River. Many farmers in this same region flood their fields, which attract waterfowl from October through March, and lease land to duck hunters.

3.19.4. O88 is situated on the Pacific flyway and is within close proximity to wintering areas for very large populations of ducks and significant populations of geese. Migratory and diurnal flight activities by birds create a substantial risk to flight operations.

3.19.5. Travis Aero Club pilots will be responsible for all wildlife hazard avoidance. Standard Bird/Wildlife Aircraft Hazard (BASH) procedures are not applicable at KO88 since it is under FAA and local municipal authority. Aero Club pilots are reminded as a rule, significant bird activity can be expected from one hour before and after sunrise/sunset. Flight operations during this time period should be avoided, especially during the spring and fall migrations.

3.19.6. Prior to departure, Travis Aero Club pilots should check the AHAS web page at: **http://www.usahas.com** to determine bird activity around military fields or training routes along their expected route of flight. Pilots should also contact local airfield management at departure and destination airfields to assess actual bird activity.

3.19.7. Prior to departure, Travis Aero Club pilots will observe runway/airfield for wildlife activity which may pose a threat during takeoff or departure. Report wildlife activity to Aero Club Management and request assistance with wildlife dispersal if needed using the UNICOM frequency. Use the standard AMC BWC listed in AFI 91-212 AMC Supp. as a guide to determine the current bird threat.

3.19.8. Travis Aero Club management will notify inbound aircraft of known significant wildlife activity on or near the airfield posing a threat to arriving aircraft using the UNICOM frequency.

3.19.9. All damaging and non-damaging wildlife strikes will be reported to the 60 AMW/SEF on an AF Form 853 and will include remains if available.

3.19.10. On occasion Travis Aero Club aircraft may receive approval to conduct transition training or full stop landings at Travis AFB. Aero Club aircraft will follow procedures and operating restrictions contained within this instruction when operating at Travis AFB.

Chapter 4

WILDLIFE HAZARD MITIGATION

4.1. Passive Deterrent Measures.

4.1.1. Long-term in nature and involve managing the local airfield environment to eliminate or reduce conditions wildlife find attractive.

4.1.2. Airfield Vegetation. Vegetative cover shall be maintained IAW AFI 91-212, *Bird/Wildlife Aircraft Strike Hazard (BASH) Management Program* section 3.2.1.2. The vegetation management guidelines outlined in sections 3.2.1.2 will be applied to the area within the WEZ which includes the banks of union creek and the old wastewater treatment area. In order for proper vegetation management to occur in these areas they must be cleared of spoils piles and other discarded materials.

4.1.2.1. Maintaining the recommended vegetation height during the rainy season (Oct-Apr) can be a challenge due to the saturated soil limiting mowing equipment access. The long-term solution to this problem is to improve drainage and remove areas of standing water/vernal pools. Until the drainage is improved all efforts must be taken to maintain the vegetation height during the rainy season because this is when the most wildlife is using the airfield. When determining mowing priority, the best practice is to begin mowing adjacent to runways, and finish in the infield or outer most vegetation areas.

4.1.3. Exclusion. Perimeter fences at Travis AFB, at a minimum, should be maintained at standards specified in AFI 91-212, chapter 3.2.1.3. Moreover, all gates and airfield access points will remain closed when not in use and should be maintained to exclude small mammals from the airfield. Exclusionary devices will be installed after coordination with the 60 AMW/SEF, USDA-WS, Airfield Management and other applicable agencies.

4.1.4. Managing Natural Areas. The 60 CE/NR is responsible for managing the base's natural resources. All proposed habitat modifications occurring on Travis AFB that have the potential to impact the BASH hazard will be discussed among BHWG members prior to any commitment to planning or resources to ensure they do not create wildlife hazards.

4.1.5. Abandoned structures and buildings within the WEZ act as wildlife attractants and must be removed. The old wastewater treatment plant attracts a variety of hazardous wildlife including swallows, waterfowl and blackbirds.

4.2. Active Deterrent Measures

4.2.1. Active deterrent measures will be performed by the wildlife harassment personnel. Their actions will consist of but not limited to: horns, blanks, pyrotechnics, trapping and if necessary lethal measures to reinforce other deterrents.

4.2.2. Pyrotechnics

4.2.2.1. Pyrotechnics are effective for dispersing most bird species and should be used for coyotes, deer, and other small animals. The easiest way to do this is to use Rohm pistols with Bombs and whistlers. These are small but very loud pyrotechnics shot from the pistol into flocks or near individual animals that frighten them away. Caution should be exercised during the summer months due to elevated fire risk. Refer to notification procedures for proper execution.

4.2.3. Firearms

4.2.3.1. Occasional lethal take of birds and other wildlife reinforces the other methods. Shooting one or two birds from a flock then following with a volley of pyrotechnics is a very effective strategy for bird deterrence. Methods of lethal take for protected migratory birds are outlined in the USFWS Migratory Bird Depredation Permit held by the 60 AMW FSO. Refer to notification procedures for proper execution.

4.2.4. Trapping

4.2.4.1. Raptor (e.g., hawks, falcons and owls) can be trapped with, but not limited to, Swedish Goshawk traps and bal-chatri traps for relocation off of Travis AFB property. USDA-WS personnel have an agreement with Sonoma Regional Parks that allows them to relocate raptors at their property. The recommended distance of relocation is at least 50 miles from Travis AFB property. Recommendations and guidance for raptor trapping and relocation can be found in the Travis AFB depredation permit issued by USFWS.

4.2.4.2. Due to more effective methods of reducing small mammal populations (rodenticides, fumigation) being prohibited on sections of the base, as well as Threatened and Endangered species concerns, trapping with cage traps is often used. Small mammals attract predators so reducing their numbers is a crucial tool in BASH management. Small mammal trapping should only be used when temperatures do not exceed a dangerous level and when precipitation is not expected. Traps should be checked at least every 24 hours. All target small mammal species caught must be euthanized on sight. Relocation of wildlife without proper permits is a violation of California state law (California Code of Regulations, Title 14, 465.5).

4.2.4.3. Trapping blackbirds during the fall and winter when they are in large flocks is highly recommended. Blackbird traps are large static traps that can trap up to hundreds of birds a day. These traps must be checked at least every 24 hours to ensure no non-target birds (Tri-colored blackbirds, raptors) have been captured.

4.3. Notification Procedures

4.3.1. Before beginning work on the airfield with the intent to performing BASH mitigation notify 60 SFS, 60 AMW/CP and 60 CE Fire Protection (60 CE/FP) of Pyrotechnic / live round use.

4.3.1.1. -60 AMW/CP: 424-5517

4.3.1.2. -60 SFS: 424-2800

4.3.1.3. -60 CE/FP: 424-3886

4.3.2. Upon entering the airfield, wildlife harassment personnel will notify 60 OSS/OSAA and ATC on the ramp frequency of bird dispersal activities before using any pyrotechnics or ammunition.

4.3.3. Once ATC has been provided initial notification of pyrotechnic use, it will not be required again while the user remains on the airfield. After exiting the airfield, wildlife harassment personnel must notify Airfield Management and ATC when bird dispersal activities are complete. NOTE: ATC approval shall be obtained prior to discharge of pyrotechnic or ammunition operations above 300' or any pyrotechnic use inside the Controlled Movement Area (CMA).

4.3.4. Prior to conducting dispersal operations affecting aircraft runway usage, the ATC will be contacted for movement coordination. Personnel performing dispersal will inform ATC of the location of birds and anticipated movement direction.

4.3.5. ATC will notify aircraft of bird dispersal operations via ATIS upon initial notification by wildlife harassment personnel to ensure pilot situational awareness. This will be broadcast until wildlife harassment personnel sign off the airfield for the day. "BIRD DISPERSAL OPERATIONS IN EFFECT AND MAY INCLUDE USE OF PYROTECHNICS."

4.4. Depredation

4.4.1. Depredation is the last resort for managing BASH threats and should only be performed after the animal no longer responds to harassment.

4.4.2. Anyone conducting depredation activities must make sure they are listed on the USFWS Migratory Bird Depredation Permit held by the 60 AMW/SEF office. Personnel who conduct depredation activities must be trained by USDA-WS personnel in proper animal identification and airfield depredation techniques.

4.4.3. Before performing depredation, the following checklist must be followed.

4.4.3.1. If harassment does not work call 60 AMW/SEF for Depredation Approval:

4.4.3.1.1. 707-424-1119

4.4.3.1.2. Approval Date and Time _____

4.4.3.2. If approval is obtained from 60 AMW/SEF offic, next gain approval from ATC over the ramp frequency.

4.4.3.3. Record the Lethal Take:

4.4.3.3.1. Name of Shooter and Observer

4.4.3.3.2. Date and Local Time

4.4.3.3.3. Location (be as specific as possible)

4.4.3.3.4. Circumstances Requiring Lethal Take

4.4.3.3.5. Methods of Harassment Used Prior to Take.

COREY A. SIMMONS, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 13-213, *Airfield Management*

AFI 31-207, *Arming and Use of Force by Air Force Personnel*

AFI 32-1053, *Pest Management Program*

AFI 32-7086, *Hazardous Materials Management*

AFI 91-202, *The US Air Force Mishap Prevention Program*

AFI 91-202 AMC Sup 1, *The US Air Force Mishap Prevention Program*

AFI 91-204, *Safety Investigations and Reports*

AFI 91-212, *Bird Aircraft Strike Hazard (BASH) Management Techniques*

AFM 31-229, *USAF Weapons Handling Manual*

AFM 91-223, *Aviation Safety Investigations and Reports*

UFC 3-260-01, *Airfield and Heliport Planning and Design*

Abbreviations and Acronyms

AFI—Air Force Instruction

AFM—Air Force Manual

AGL—Above Ground Level

AMC—Air Mobility Command

AHAS—Avian Hazard Advisory System

ASR—Airfield Surveillance Radar

ATC—Air Traffic Control

ATIS—Airport Terminal Information System

BAM—Bird Avoidance Model

BASH—Bird Aircraft Strike Hazard

BHWG—Bird Hazard Working Group

BWC—Bird Watch Condition

CES—Civil Engineering Squadron

DER—Departure End of Runway

EPA—Environmental Protection Agency

FAA—Federal Aviation Administration

IAW—In Accordance With

IMC—Instrument Meteorological Conditions

OPR—Office of Primary Responsibility

RAPCON—Radar Approach Control

AFSAS—Air Force Safety Automated System

SFS—Security Forces Squadron

TACC—Tanker Airlift Control Center

UFC—Unified Facilities Criteria

USDA—United States Department of Agriculture

VMC—Visual Meteorological Conditions